

# Heterogeneous Households and Aggregate Dynamics: Belief-Invariance and Data Inversion<sup>1</sup>

## Online Appendix

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# A Proofs

## A.1 Proof of Proposition 1

**Proposition 1** (Analytical household policies and wealth dynamics).

*The optimal consumption and labor supply policies are:*

$$c(x; X) = \frac{1 - \beta}{1 + \theta} (1 + r(X)(1 - \tau)e)a + \left( \frac{1 - \beta}{1 + \theta} a\xi + \kappa \right) w(X)(1 - \tau)Z_L z \quad (1)$$

$$n(x; X) = \max \left\{ \kappa(1 + \theta) + \xi a - \frac{\theta c(x; X)}{w(X)(1 - \tau)Z_L z}, 0 \right\} \quad (2)$$

*The household's chosen saving is  $\hat{a}'(x; X) = \beta a ((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z\xi)$ .*

*The actual law of motion for wealth, after the unanticipated government transfer, is:*

$$a'(x; X) = \hat{a}'(x; X) + T(X) = \beta a ((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z\xi) + T(X). \quad (3)$$

*Proof.*

I assume the constraint  $a' \geq 0$  does not bind and later verify that the solution satisfies this. The household is unaware of the transfer  $T$  and optimizes based solely on factor income. From the first-order conditions,

$$[c] : \frac{1}{c - \kappa w(X)(1 - \tau)Z_L z} = \lambda \quad (4)$$

$$[a'] : \beta \mathbb{E}[v_a(x'; X')] = \lambda \quad (5)$$

$$[n] : \frac{\theta}{\kappa + \xi a - n} = \lambda w(X)(1 - \tau)Z_L z. \quad (6)$$

From the envelope condition,

$$[a] : v_a(x; X) = \frac{\theta \xi}{\kappa + \xi a - n} + \lambda(1 + r(X)(1 - \tau)e). \quad (7)$$

From Equations (4) and (6),

$$\frac{\theta}{\kappa + \xi a - n} = \frac{w(X)(1 - \tau)Z_L z}{c - \kappa w(X)(1 - \tau)Z_L z}. \quad (8)$$

Rearranging,

$$n(x; X) = \kappa(1 + \theta) + \xi a - \theta \frac{c(x; X)}{w(X)(1 - \tau)Z_L z}. \quad (9)$$

The budget constraint can be written as:

$$(1 + \theta)(c - \kappa w(X)(1 - \tau)Z_L z) + a' = ((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z \xi)a. \quad (10)$$

Conjecture  $v(a, z, e; X) = A(z, e, X) \log a + B(z, e, X)$ . Under log utility, the Euler equation combined with the conjectured form yields  $a' = \beta a R_i$ , where  $R_i = (1 + \tilde{r}e + \tilde{w}Z_L z \xi)$ —the standard result of [Brock and Mirman \(1972\)](#). The government then adds the unanticipated transfer:  $a'_{\text{actual}} = \beta a R_i + T$ . Substituting back confirms  $A(z, e, X) = (1 + \theta)/(1 - \beta)$ , and  $B$  satisfies a well-defined recursion. The coefficient  $A$  is independent of beliefs about future aggregates, which is the basis for Theorem 1; see Appendix [A.8](#) for the general  $\zeta$  recursion when  $\beta$  varies. Finally, the household's chosen saving  $\hat{a}' = \beta a R_i$  is non-negative for every  $a \geq 0$ , since  $\beta \in (0, 1)$  and  $R_i \geq 0$  under the maintained assumption  $1 + \tilde{r}e \geq 0$ ; so the borrowing constraint  $a' \geq 0$  is slack in the optimization itself, independent of  $T$ . ■

## A.2 Proof of Proposition 2

*Proof.* Write  $s(a) = (\alpha a + T)/(\gamma a + K_d + T)$ , where

$$\alpha \equiv \beta(1 + \tilde{r}e + \tilde{w}Z_L z \xi) - 1, \quad \gamma \equiv \frac{\tilde{w}Z_L z \xi(1 + \theta\beta)}{1 + \theta} - \frac{\theta(1 - \beta)(1 + \tilde{r}e)}{1 + \theta} + \tilde{r}e, \quad K_d \equiv \tilde{w}Z_L z \kappa \theta.$$

Property (ii):  $s(0; X) = T/(K_d + T) > 0$  since  $T > 0$  and  $K_d > 0$ . Property (i):  $\lim_{a \rightarrow \infty} s(a; X) = \alpha/\gamma$ , the limit of the fractional-linear function in  $a$ . Property (iii): direct differentiation gives

$$\frac{\partial s}{\partial a} = \frac{\alpha(K_d + T) - \gamma T}{(\gamma a + K_d + T)^2}, \quad \frac{\partial^2 s}{\partial a^2} = \frac{-2\gamma[\alpha(K_d + T) - \gamma T]}{(\gamma a + K_d + T)^3}.$$

The numerator  $\Delta \equiv \alpha(K_d + T) - \gamma T$  is positive when two conditions hold: (a)  $\alpha > 0$ , which is the dynamic-efficiency condition  $\beta R_i > 1$ ; and (b)  $K_d$  dominates the transfer correction  $-(\gamma - \alpha)T$ . Both conditions are satisfied throughout the parameter region examined in the quantitative section. Specifically,  $\beta R_i > 1$  in every period of the historical inversion, since  $\tilde{r} + \tilde{w}Z_L \xi > (1 - \beta)/\beta \approx 0.04$ . The transfer is small relative to subsistence:  $T \approx 0.001K \ll K_d$  at the calibrated  $\eta$ . Hence  $\Delta > 0$  and  $\gamma > 0$ , giving  $\partial s/\partial a > 0$  and  $\partial^2 s/\partial a^2 < 0$ . ■

## A.3 Proof of Proposition 3

*Proof.* The saving policy  $a'(x; X) = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L \xi z) + T$  is linear in  $a$  with coefficients that depend on  $(z, e)$  and current prices. Aggregating across the distribution:

For aggregate capital:  $K' = \int a'(x; X) d\Phi = \beta[(1+\tilde{r}+\tilde{w}Z_L\xi)K + \tilde{r}\text{cov}(e, a) + \tilde{w}Z_L\xi\text{cov}(z, a)] + T$ , using  $\mathbb{E}[e] = \mathbb{E}[z] = 1$ .

For part (i) (contemporaneous sufficiency): Proposition 1 shows that individual policies  $c(x; X)$ ,  $a'(x; X)$ ,  $n(x; X)$  are linear in  $a$  with coefficients depending on  $(z, e)$  and prices  $(r, w)$ . The labor supply equation depends on  $\Phi$  only through  $(K, \text{cov}(z, a), \text{cov}(e, a))$ , so prices are functions of  $\mathcal{M}$  alone. Hence all equilibrium allocations are determined by  $\mathcal{M}$  and the aggregate state.

For part (iii) (factor prices and aggregate labor): direct aggregation of the individual policies using  $\mathbb{E}[e] = \mathbb{E}[z] = 1$  yields the stated expressions.

*Unique existence of aggregate labor.* The aggregate labor equation (20) is implicit in  $L$  because the after-tax wage  $\tilde{w} = (1 - \tau_L)(1 - \alpha)A(K_{eff}/L)^\alpha$  depends on  $L$ . Define  $g(L) \equiv L - \text{RHS}(L)$  where RHS is the right-hand side of (20) evaluated at prices consistent with  $L$ . The first term  $(\kappa + Z_L\xi \cdot f(\beta, \theta) \cdot (K + \text{cov}(z, a)))$  is independent of  $L$ . The second term,  $-\theta(1-\beta)/((1+\theta)\tilde{w}Z_L) \cdot W$  where  $W = K(1+\tilde{r}) + \tilde{r}\text{cov}(e, a)$ , depends on  $L$  through  $\tilde{w} \propto L^{-\alpha}$  and  $\tilde{r} \propto L^{1-\alpha}$ . Since  $\alpha \in (0, 1)$ ,  $\tilde{w}$  is strictly decreasing in  $L$  while  $W/\tilde{w} \propto L^\alpha \cdot L^{1-\alpha} = L$ . Hence the negative term scales as  $-c \cdot L$  for some  $c > 0$ , making  $g'(L) = 1 + c > 0$  strictly positive. Continuity and strict monotonicity of  $g$  guarantee a unique zero. ■

*Remark on distributional dynamics.* For  $\text{cov}(z', a')$ : expanding  $\mathbb{E}[z'a'] - \mathbb{E}[z']\mathbb{E}[a']$  using  $z' = (1 - \rho_z) + \rho_z z + \varepsilon'_z$  and  $a' = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L\xi z) + T$  yields terms involving  $\mathbb{E}[\hat{z}^2 a]$  and  $\mathbb{E}[\hat{z}\hat{e}a]$  (where  $\hat{z} = z - 1$ ,  $\hat{e} = e - 1$ ). These three-way cross-moments are not, in general, functions of  $\mathcal{M}$  alone. Under joint normality of  $(\hat{z}, \hat{e}, a)$ , the Isserlis (Wick) theorem gives  $\mathbb{E}[\hat{z}^2 a] = \sigma_z^2 K$  and  $\mathbb{E}[\hat{z}\hat{e}a] = 0$  (using independence of  $z$  and  $e$ ), and the covariance dynamics close into a  $3 \times 3$  linear recursion. However, the Kesten-process wealth dynamics generate Pareto tails that violate normality. Therefore, tracking the full distribution via agent-based simulation is necessary for the forward evolution of the covariances, while the  $K'$  equation and all contemporaneous allocations remain exact. ■

## A.4 Proof of Corollary 2

*Proof.* The individual law of motion  $a' = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L\xi z) + T$  is a Kesten process with multiplicative coefficient  $R_i = \beta((1 + \tilde{r}e) + \tilde{w}Z_L\xi z)$  and additive intercept  $T > 0$ . By Kesten (1973), when  $\mathbb{E}[\log R_i] < 0$  and  $T > 0$ , the process admits a unique stationary distribution with Pareto tails:  $\Pr(a > x) \sim x^{-\zeta}$  for large  $x$ , where  $\zeta$  solves  $\mathbb{E}[R_i^\zeta] = 1$ . The additive component  $T > 0$  is essential: without it, the purely multiplicative system does not admit a non-degenerate stationary solution. ■

## A.5 Proof of Corollaries 3 and 4

*Proof of Corollary 3.* Set  $e = 1$  throughout. Then  $\text{cov}(e, a) = 0$  for all  $t$ , and the  $K'$  equation reduces to  $K' = \beta[(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \tilde{w}Z_L\xi \text{cov}(z, a)] + T$ , with  $\mathcal{M} = [K, \text{cov}(z, a)]$  sufficient for all contemporaneous allocations. ■

*Proof of Corollary 4.* Set  $e = 1$  and  $z = 1$ . Then both covariances vanish identically, and the system reduces to a single equation  $K' = \beta(1 + \tilde{r} + \tilde{w}Z_L\xi)K + T$ . The aggregate consumption function follows from  $C = Y + (1 - \delta)K - K'$ . ■

## A.6 Proof of Theorem 1

*Proof.* The saving policy  $a'(x; X) = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L\xi z) + T$  depends only on current after-tax prices  $(\tilde{r}, \tilde{w})$ , current  $Z_L$ , the current transfer  $T$ , and the household's current state  $(a, z, e)$ . None of these objects involves expectations of future aggregate states. The consumption policy  $c(x; X)$  and labor supply  $n(x; X)$  inherit this property. Since all individual policies are independent of beliefs about future aggregates, the aggregate law of motion—including  $K'$  from the  $K'$  equation in the main text (eq. (17)) and the distributional evolution of  $\Phi$ —is also independent of beliefs: it depends only on current prices (which are functions of the current state  $\mathcal{M}$ ) and the current transfer.

The mechanism is the certainty-equivalence property of log preferences: income and substitution effects of future return changes exactly cancel, so saving depends only on current resources and prices. This aggregates because linear policies preserve the moment structure. ■

## A.7 Proof of Proposition 4

*Proof.* The BRCE requires that household policies be optimal given *arbitrary* beliefs about aggregate dynamics. Theorem 1 establishes that the saving policy is belief-invariant. Since the aggregate law of motion depends only on the collection of saving policies (through the moment aggregation), and since these policies are identical regardless of beliefs, the BRCE law of motion coincides with the MRCE law of motion. Uniqueness of the MRCE then implies uniqueness of the BRCE, and the allocations are identical. ■

## A.8 Saving share under time-varying $\beta$

*Derivation.* Since  $\beta$  is an element of the aggregate state  $X$  and may vary over time, the Bellman equation for the multiplicative component of wealth (ignoring the additive transfer

$T$ ) is:

$$v_t(a, z, e; X) = \max_{a'} (1 + \theta) \log \tilde{c} + \beta_t \mathbb{E}_t [v_{t+1}(a', z', e'; X')], \quad (11)$$

where  $\tilde{c} \equiv c - \kappa w(1 - \tau)Z_L z$  is consumption net of the subsistence floor. Conjecture  $v_t(a, z, e; X) = \zeta_t \log a + B_t(z, e, X)$ , where  $\zeta_t$  may depend on aggregate conditions but not on individual wealth.

From the envelope condition,  $v_{a,t} = \zeta_t/a$ . Combined with the first-order conditions (as in Appendix A.1), the envelope condition yields  $\zeta_t/a = \lambda_t \Pi_t$ , where  $\Pi_t = (1 + \tilde{r}_t e) + \tilde{w}_t Z_L \xi z$  is the gross multiplicative return on wealth. Since  $\lambda_t = 1/\tilde{c}_t$ , this gives  $\tilde{c}_t = a \Pi_t / \zeta_t$ , and the budget constraint implies:

$$a' = \Pi_t a \left(1 - \frac{1 + \theta}{\zeta_t}\right). \quad (12)$$

The first-order condition for  $a'$  requires  $\beta_t \mathbb{E}_t [\zeta_{t+1}/a'] = \zeta_t/(a \Pi_t)$ . Substituting the expression for  $a'$  and simplifying:

$$\zeta_t = (1 + \theta) + \beta_t \mathbb{E}_t [\zeta_{t+1}]. \quad (13)$$

The saving share out of the multiplicative component is therefore:

$$s_t = 1 - \frac{1 + \theta}{\zeta_t} = \frac{\beta_t \mathbb{E}_t [\zeta_{t+1}]}{\zeta_t}. \quad (14)$$

**Constant  $\beta$ .** When  $\beta$  is constant,  $\zeta = (1 + \theta)/(1 - \beta)$  and  $s = \beta$ , recovering the steady-state result.

**Belief invariance.** Under the BRCE, households treat the current  $\beta$  as permanent. The recursion (13) then gives  $\zeta = (1 + \theta)/(1 - \beta)$ , so  $s = \beta$ . This is a static function of the current discount factor—no expectations are required. The saving policy is  $a'(x; X) = \beta a \Pi + T$ , and the aggregate capital law of motion  $K' = \beta[(1 + \tilde{r} + \tilde{w} Z_L \xi)K + \tilde{r} \text{cov}(e, a) + \tilde{w} Z_L \xi \text{cov}(z, a)] + T$  inherits this belief invariance.

**Time-varying  $\tau_t$ .** The tax rate enters only through after-tax prices in the income base  $\Pi_t$ , and through the transfer  $T_t$  (which is an MIT shock). Since  $\tau_t$  does not appear in the  $\zeta$  recursion (13), time-varying flat tax rates preserve the analytical structure without any additional state variables.

**Separate capital and labor tax rates.** The same logic extends to separate time-varying tax rates  $\tau_{K,t}$  and  $\tau_{L,t}$ . After-tax factor prices become  $\tilde{r}_t = r_t(1 - \tau_{K,t})$  and  $\tilde{w}_t = w_t(1 - \tau_{L,t})$ , and the transfer is  $T_t = \eta[\tau_{K,t}(r_t + \delta)K_t + \tau_{L,t} w_t L_t]$ . Because both rates

enter multiplicatively on factor incomes, the household's next-period wealth  $a' = s_t \cdot a\Pi_t + T_t$  remains *linear* in individual wealth  $a$ —the key property that preserves the analytical structure. In particular, the saving share  $s_t = \beta_t$  is still independent of  $(\tau_{K,t}, \tau_{L,t})$ : tax rates shift *what* is saved (through  $\Pi_t$  and  $T_t$ ) but not the *share* of wealth saved. Progressive taxes or tax brackets, by contrast, would make the effective marginal rate depend on household-level income  $a\Pi_t$ , breaking linearity in  $a$  and destroying the MRCE's closed-form aggregation.

■

## B Extended Theoretical Discussion

### B.1 Transition matrix interpretation

The MRCE state vector has three components: aggregate capital  $K$  (the mean of the wealth distribution),  $\text{cov}(z, a)$  (the alignment between labor productivity and wealth), and  $\text{cov}(e, a)$  (the alignment between return heterogeneity and wealth). The first is the single state that [Krusell and Smith \(1998\)](#) found nearly sufficient numerically.

The aggregate capital law of motion  $K' = \beta[(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \tilde{r}\text{cov}(e, a) + \tilde{w}Z_L\xi\text{cov}(z, a)] + T$  shows how the two covariances interact with capital. The  $\tilde{r}\text{cov}(e, a)$  term captures return-wealth sorting: high-return wealthy households save more than average returns and average wealth would predict. The  $\tilde{w}Z_L\xi\text{cov}(z, a)$  term captures productivity-wealth sorting through wealth-leisure complementarity. The two sorting channels operate independently (because  $z$  and  $e$  are independent), connected only through their common dependence on  $K$ .

### B.2 Signal-to-noise ratios

The signal-to-noise ratios  $\phi_z \equiv \sigma_z^2/(\sigma_z^2 + \sigma_{\varepsilon_z}^2)$  and  $\phi_e \equiv \sigma_e^2/(\sigma_e^2 + \sigma_{\varepsilon_e}^2)$  control the rate at which idiosyncratic shocks reshuffle households across the distribution. High  $\phi$  (close to one) means persistent shocks and slow reshuffling: sorting between productivity and wealth strengthens over time. Low  $\phi$  (close to zero) means rapid reshuffling: the covariance between productivity and wealth dissipates quickly. Economies with persistent idiosyncratic shocks exhibit slow-moving distributional states that interact strongly with aggregate fluctuations; economies with transitory shocks have fast-adjusting distributions that play a smaller aggregate role.

### B.3 Special cases: extended discussion

**Approximate aggregation** When there is no return heterogeneity ( $e = 1$ ), the system reduces to two states:  $K$  and  $\text{cov}(z, a)$ . This is the analytical counterpart of the Krusell–



Smith economy. [Krusell and Smith \(1998\)](#) found numerically that the mean of the wealth distribution is “nearly sufficient” to forecast prices. The present result shows that the mean is *exactly* sufficient if and only if  $\text{cov}(z, a) = 0$ . In practice, this covariance is small relative to  $K$  in many calibrations, explaining why approximate aggregation works well numerically. But the error grows with the persistence and dispersion of idiosyncratic productivity—precisely the conditions under which the Krusell–Smith approximation breaks down.

**Representative-agent nesting** When all heterogeneity is eliminated, both covariances vanish and the model reduces to a stochastic growth model with log preferences and endogenous labor. The aggregate consumption function  $C = \frac{1-\beta}{1+\theta}(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \kappa\tilde{w}Z_L$  has a permanent-income term proportional to wealth and a subsistence term proportional to the wage. The heterogeneous-agent model adds distributional corrections that, while small in normal times, become significant during episodes of high inequality.

## B.4 State-dependent inequality dynamics

The  $K'$  equation makes inequality feedback precise. Higher capital raises returns, which increases  $\tilde{r} \text{cov}(e, a)$  (the return-sorting contribution to aggregate saving), which raises future capital further. Similarly, higher capital raises wages, which increases  $\tilde{w}Z_L\xi \text{cov}(z, a)$  (the productivity-sorting contribution). This self-reinforcing loop generates endogenous amplification: a positive shock raises capital, which strengthens sorting, which raises future capital.

This “wealth spiral” is consistent with the secular rise in top wealth shares documented by [Piketty \(2014\)](#) and [Saez and Zucman \(2016\)](#). The rate of concentration is governed by the persistence of the distributional dynamics, measured by the dominant eigenvalue in the agent-based simulation. The steady-state level is pinned down by the balance between multiplicative compounding ( $R_i$ ) and the additive transfer  $T$ . Two forces limit concentration. Consumption:  $\beta < 1$  ensures that a fraction  $1 - \beta$  of wealth is consumed each period. The transfer  $T$  provides an additive floor that benefits low-wealth households disproportionately.

## C Asset Pricing Details

The model delivers a novel asset pricing result: individual return heterogeneity, traditionally viewed as diversifiable, carries a positive risk premium because it has aggregate consequences ([Constantinides and Duffie, 1996](#); [Storesletten et al., 2004](#)).

In the standard complete-markets framework, idiosyncratic risk washes out by the law

of large numbers. Here it does not, because return heterogeneity creates systematic wealth sorting: households with persistently high  $e$  accumulate more wealth, work less through the wealth effect, and thereby depress aggregate labor supply and returns for *everyone*. Because  $e$  affects aggregate outcomes through  $\text{cov}(e, a)$ , it is not fully diversifiable.

## C.1 Static and dynamic decomposition

The asset pricing implications decompose into a static and a dynamic component.

**Static effect** Holding the aggregate state fixed, higher  $\text{cov}(e, a)$  reduces the current return:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \text{cov}(e, a)} = \frac{\partial r}{\partial L} \frac{\partial L}{\partial \text{cov}(e, a)} < 0 \quad (15)$$

Higher sorting strengthens the wealth effect on labor supply, reducing employment and the marginal product of capital.

**Dynamic effect** One period ahead, the effect is reinforced:

$$\frac{\partial r(\mathcal{M}', A')}{\partial \text{cov}(e, a)} = \frac{\partial r'}{\partial K'} \frac{\partial K'}{\partial \text{cov}(e, a)} + \frac{\partial r'}{\partial \text{cov}(e', a')} \frac{\partial \text{cov}(e', a')}{\partial \text{cov}(e, a)} < 0 \quad (16)$$

Both terms are negative: higher sorting today raises capital accumulation (depressing future returns through diminishing marginal products) and propagates into tomorrow's sorting (continuing to depress returns through labor supply).

## C.2 Wealth mobility

The model provides an analytical half-life of wealth:

$$T(\mathcal{M}, A) = \frac{\log(2)}{\log(\beta(1 + r(\mathcal{M}, A)(1 - \tau)))} \quad (17)$$

Faster mean reversion of  $\text{cov}(e, a)$  implies quicker distributional adjustment, but slow rank mobility means individual wealth positions change gradually. The combination generates persistent aggregate effects of idiosyncratic return shocks. These results suggest that models with heterogeneous returns (Fagereng et al., 2020; Bach et al., 2020) have implications for asset prices and aggregate dynamics, not just the wealth distribution.

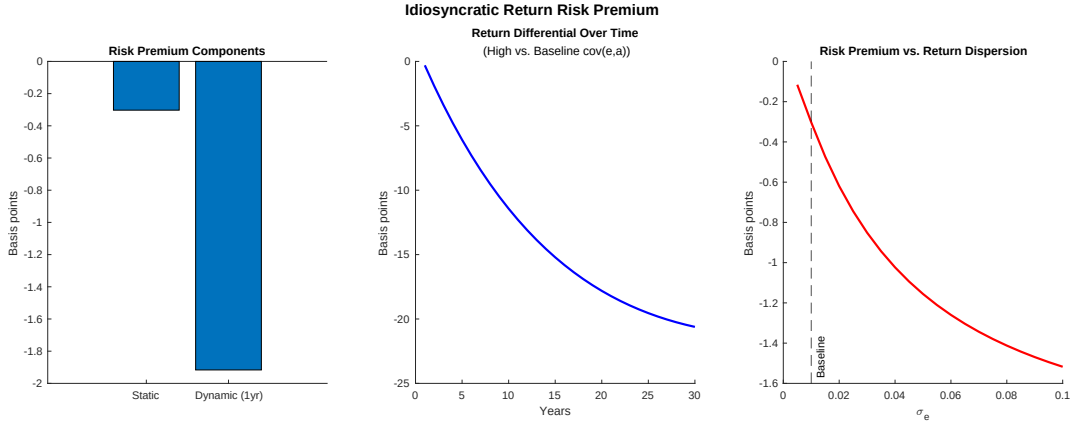
### C.3 Risk premium sensitivity

To quantify the pricing channel, we compare steady-state economies at different  $\sigma_e$ . At baseline ( $\sigma_e = 0.01$ ), the interest rate effect is  $-0.001$  basis points—negligible. The premium grows quadratically:

$$\underbrace{-0.03 \text{ bp}}_{\sigma_e=0.05} \quad \underbrace{-0.10 \text{ bp}}_{\sigma_e=0.10} \quad \underbrace{-0.70 \text{ bp}}_{\sigma_e=0.50} \quad \underbrace{-1.01 \text{ bp}}_{\sigma_e=0.83}$$

The threshold for a 1 basis point premium requires  $\sigma_e \approx 0.83$ . At empirically realistic  $\sigma_e$ , the main consequence of return heterogeneity operates through the *amplification channel* (state-dependent impulse responses), not the *pricing channel*. Figure C.1 visualizes these results.

Figure C.1: Return risk premium sensitivity



Notes: Left: static vs. dynamic premium components at steady state. Center: cumulative return differential over 30 years. Right: premium as a function of return dispersion  $\sigma_e$ .

## D Epstein-Zin Extension

Under Epstein-Zin preferences with unit elasticity of intertemporal substitution (EIS = 1), the flow composite is:

$$\tilde{c} \equiv c - \kappa Z_L z w(X)(1 - \tau), \quad m \equiv (\kappa + \xi a) - n, \quad X^{\text{comp}} \equiv \tilde{c} m^\theta.$$

The intratemporal optimality condition implies  $m = (\theta/(w(X)(1 - \tau)Z_L z))\tilde{c}$ , so:

$$X^{\text{comp}} = \tilde{c}^{1+\theta} \left( \frac{\theta}{w(X)(1 - \tau)Z_L z} \right)^\theta.$$

The recursive value function satisfies:

$$\log V_t = (1 - \beta) \log X_t + \beta \log \left( \mathbb{E}_t [V_{t+1}^{1-\gamma}] \right)^{\frac{1}{1-\gamma}}, \quad \beta \in (0, 1), \quad \gamma > 0.$$

The critical result is that, with EIS = 1, the optimal saving share  $s_a(s)$  satisfies the same type of fixed-point condition as in the log case:

$$s_a(s) = \left[ 1 + \left( \frac{\beta(1 + \theta)^p \bar{B}(s)}{C(s)} \right)^{\frac{1}{p-1}} \right]^{-1},$$

where  $p \equiv (1 + \theta)(1 - \sigma)$ ,  $C(s) \equiv (\theta/(w(X)(1 - \tau)Z_L z))^{\theta(1-\sigma)}$ , and  $\bar{B}(s) \equiv \mathbb{E}[B(s') \mid s]$  is a conditional expectation computed from the fixed-point system:

$$B(s) = \Pi(s)^p \left[ C(s) \left( \frac{1 - s_a(s)}{1 + \theta} \right)^p + \beta s_a(s)^p \bar{B}(s) \right],$$

with  $\Pi(s) \equiv (1 + r(X)(1 - \tau))e + w(X)(1 - \tau)Z_L z \xi$ .

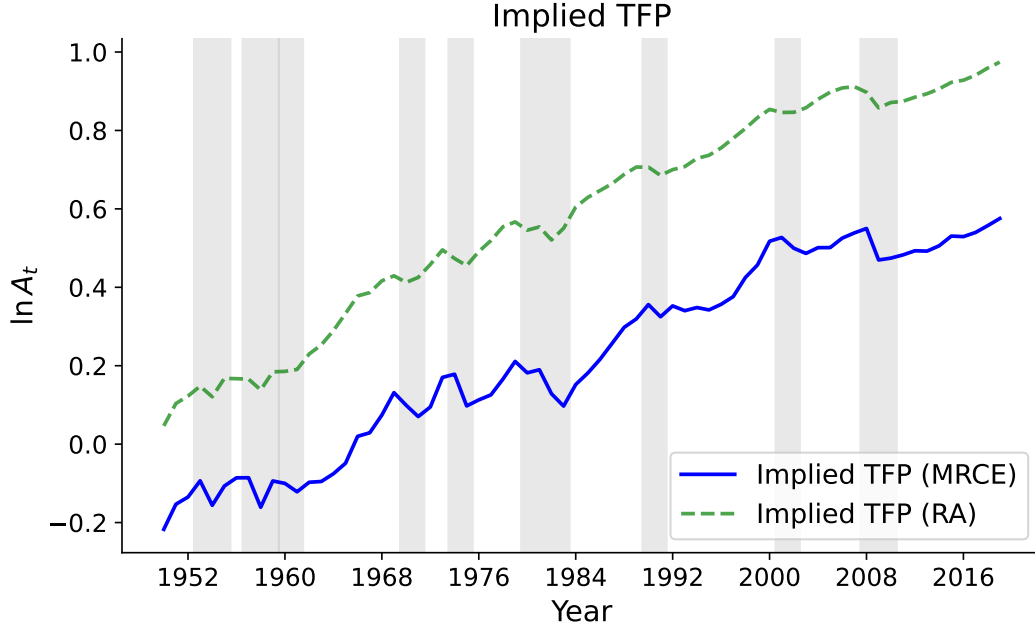
The MRCE and BRCE continue to hold under Epstein-Zin preferences with unit EIS. The saving share remains independent of wealth, and the law of motion retains its closed-form matrix structure. The parameter  $\gamma$  controls risk aversion over consumption *lotteries* (cross-sectional), while  $\beta$  and  $\theta$  control intertemporal and intratemporal tradeoffs (time-series). This separation allows the model to match both the equity premium (through  $\gamma$ ) and the saving rate (through  $\beta$ ). With  $\gamma \gg 1$ , the wealth-sorting mechanism generates a substantially larger risk premium than under log utility.

## E Additional Quantitative Results

### E.1 Implied TFP comparison

Figure E.1 compares implied TFP from the MRCE (accounting for distributional covariances) with TFP from a representative-agent version (all covariances set to zero). The two series track closely at business cycle frequencies, but the HA model implies a slightly different level, reflecting distributional sorting's contribution to labor supply. The Fernald utilization-adjusted TFP series serves as an external benchmark.

Figure E.1: Implied TFP comparison



*Notes:* Implied TFP from the MRCE compared to the representative-agent analogue and the Fernald utilization-adjusted TFP series. Shaded areas indicate NBER recession dates.

## E.2 Structural interpretation of $Z_{L,t}$

The aggregate labor productivity  $Z_{L,t}$ , introduced as a model primitive in Section 2, is a structural object—not a reduced-form residual. Three properties distinguish it from reduced-form labor wedges.

**Model primitive with distributional consequences**  $Z_{L,t}$  scales household effective earnings  $\tilde{w}_t Z_{L,t} z$  without entering the production function as a separate factor (it enters only through the efficiency-weighted aggregate  $L_t \equiv Z_{L,t} \int z n d\Phi$ , as discussed below). It appears in three places in the equilibrium:

1. The effective wage in household labor supply:  $\tilde{w}_t Z_{L,t}$ ,
2. The wealth-leisure complementarity:  $\tilde{w}_t Z_{L,t} \tilde{\xi}_t$ , where  $\tilde{\xi}_t \equiv \xi / A_t^{1/(1-\alpha)}$  is the growth-adjusted complementarity coefficient that holds the wealth-effect strength fixed in detrended units, and
3. The saving policy, which governs the evolution of the wealth distribution.

Through channel (3), a decline in  $Z_{L,t}$  reduces the rate at which productivity-wealth sorting accumulates, altering the dynamics of wealth inequality. This distinguishes  $Z_{L,t}$  from a [Chari](#)

et al. (2007) labor wedge, which operates only through the intratemporal margin and has no distributional content.

**Identification through effective labor input**  $Z_{L,t}$  enters the production function only through  $L_t \equiv Z_{L,t} \int z n(x; X) d\Phi$  (effective labor input), not as a separate factor multiplying  $L^{1-\alpha}$ . This rules out the labor-augmenting interpretation: a shift in  $Z_{L,t}$  rescales effective hours one-for-one without changing the marginal-product schedule of  $L$  given  $L$ . In the inversion (main text Section 4), the observed labor series is interpreted as the model’s effective labor aggregate  $L_t$ , so that  $A_t$  is recovered period by period from  $A_t = Y_t / (K_{eff,t}^\alpha L_t^{1-\alpha})$ .  $Z_{L,t}$  is then identified residually from the household labor-supply identity given  $(A_t, \beta_t, K_t, \text{cov}_t(z, a), \text{cov}_t(e, a))$ . The asymmetry is testable: a shift in  $Z_{L,t}$  alters hours and investment composition without altering technology.

**External validation** Section 4.3 reports correlations between  $Z_{L,t}$  and labor market indicators not used in the inversion: the employment-population ratio (level  $\rho = 0.48$ , HP-cycle  $\rho = 0.81$ ), the labor force participation rate (0.28, 0.54), and average weekly manufacturing hours (0.42, 0.41). Only GDP, the capital stock, and per-capita hours enter the inversion. The strong HP-cycle correlations confirm that  $Z_{L,t}$  tracks the extensive margin of labor utilization at business-cycle frequencies—exactly the margin it is designed to capture—and the moderate level correlations align  $Z_{L,t}$  with secular movements in employment intensity, supporting its interpretation as a structural utilization-and-matching wedge.

Equally informative is the *negative* HP-cycle correlation ( $\rho = -0.70$ ) with the BLS labor composition index (MPU4900092, 1987–2019), which measures workforce quality through demographics. In recessions, low-productivity workers are disproportionately separated, raising average quality—the “cleansing” effect of Caballero and Hammour (1994). Meanwhile,  $Z_{L,t}$  declines through underutilization, matching deterioration, and foregone on-the-job learning. The sign reversal confirms that  $Z_{L,t}$  captures a *utilization* margin distinct from demographic composition.

**Comparison with the CKM labor wedge** In Chari et al. (2007), the labor wedge  $1 - \tau_L$  measures the gap between the intratemporal optimality condition and data but does not identify its source. In contrast,  $Z_{L,t}$  has a specific structural role: it is the mean of the idiosyncratic productivity distribution, and its variation maps to composition, matching, and utilization channels measurable independently. The external validation above confirms this structural content is empirically relevant, not merely a relabeling of the CKM residual.

### E.3 Aggregate dynamics

This appendix provides the full details of the aggregate dynamics results summarized in the main text (Section 5.1).

**Investment decomposition** Prediction **P1** (approximate aggregation) can be tested directly. Investment decomposes as:

$$I^{\text{rep}} = \beta(1 + \tilde{r} + \tilde{w}Z_L\xi)K + T - (1 - \delta_t)K \quad (\text{representative-agent channel}) \quad (18)$$

$$I^{\text{ret}} = \beta\tilde{r} \text{cov}(e, a) \quad (\text{return-sorting channel}) \quad (19)$$

$$I^{\text{prod}} = \beta\tilde{w}Z_L\xi \text{cov}(z, a) \quad (\text{productivity-sorting channel}) \quad (20)$$

Figure E.2 plots the three channels. The representative-agent channel accounts for 93.2 percent of mean investment, confirming analytically what Krusell and Smith (1998) found numerically. The distributional channels contribute the remaining share, yet they generate the state-dependent amplification that a representative-agent model cannot produce.

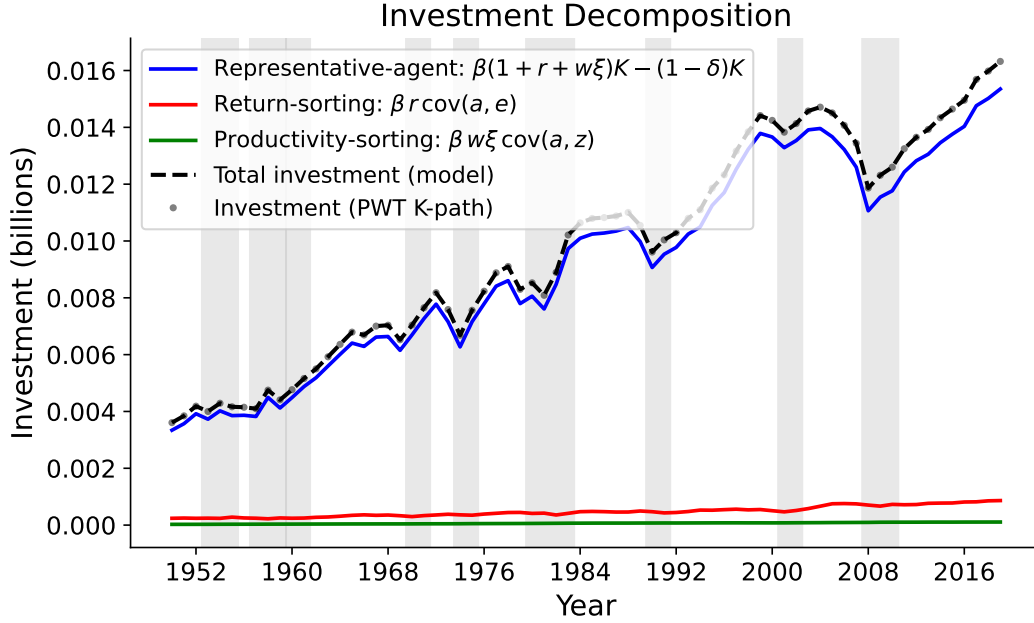
Note that model-implied investment  $I_t^{\text{model}} = K_{t+1} - (1 - \delta_t)K_t$  need not coincide with measured gross investment from the national accounts. The model abstracts from inventory investment, government investment, and residential construction included in NIPA aggregates.

**State-dependent dynamics** To test prediction **P2** (state-dependent amplification), we compute generalized impulse response functions (GIRFs) for a 2 percent negative TFP shock launched from two initial states: the 1952 distribution (low inequality,  $\text{cov}(e, a) = 0.001$ ) and the 2019 distribution (high inequality,  $\text{cov}(e, a) = 0.006$ ). Following Koop et al. (1996), each GIRF averages over 1,000 stochastic TFP paths with common innovations. The impact effect on output is 3 percent larger from 2019 ( $-2.29\%$  vs.  $-2.22\%$ ), and the cumulative 10-year loss is modestly larger ( $-20.4\%$  vs.  $-20.2\%$ ). The amplification is quantitatively small at the baseline calibration ( $\sigma_e = 0.01$ ), consistent with the negligible return risk premium.

The mechanism operates through three reinforcing channels. First, the *exposure channel*: the return-sorting term  $\beta\tilde{r} \text{cov}(e, a)$  in the capital recursion means that aggregate investment depends on the covariance between returns and wealth. When  $\text{cov}(e, a)$  is large, a larger share of saving originates from households most sensitive to the aggregate return  $\tilde{r}$ , amplifying the investment contraction.

Second, the *labor withdrawal channel*: the wealth-effect term  $-\frac{\theta(1-\beta)}{(1+\theta)\tilde{w}Z_L}\tilde{r} \text{cov}(e, a)$  shows that return-wealth sorting reduces aggregate labor supply. When  $\text{cov}(e, a)$  is large, wealthy households withdraw more labor through wealth-leisure complementarity ( $\kappa + \xi a$ ), depressing

Figure E.2: Decomposition of aggregate investment into three channels



*Notes:* The three channels are the representative-agent channel ( $I^{\text{rep}}$ ), the return-sorting channel ( $I^{\text{ret}}$ ), and the productivity-sorting channel ( $I^{\text{prod}}$ ). The representative-agent channel accounts for 93.2% of mean investment. Gray dots show measured NIPA investment for reference; the gap reflects model abstractions (inventories, government and residential investment). Shaded areas indicate NBER recession dates.

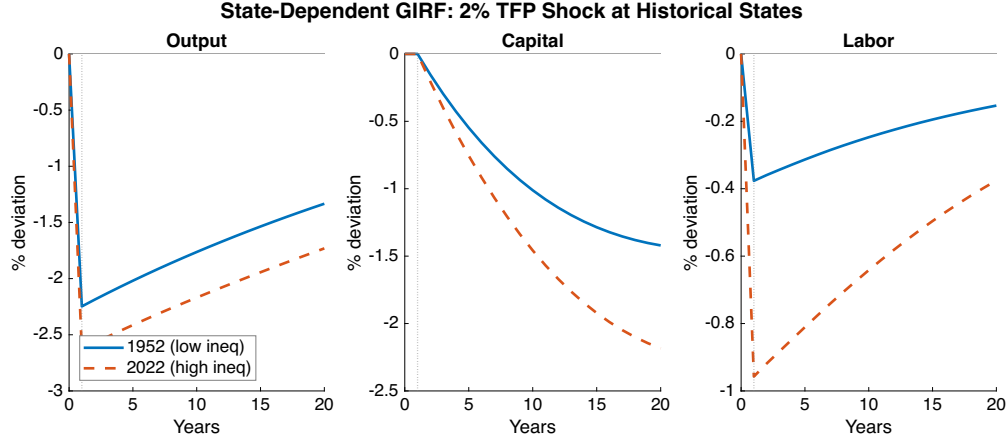
output beyond the direct capital channel.

Third, the *persistence channel*: higher capital raises returns, which feeds back into future return-wealth sorting through the individual saving policy. The distributional dynamics exhibit near-unit-root persistence along the historical path, so the distributional state does not mean-revert. Sorting that amplified the initial shock *persists*, keeping the economy on a lower trajectory. At the baseline calibration, the gap between high- and low-inequality GIRFs is small (1 percent in cumulative losses), reflecting that  $\sigma_e = 0.01$  generates only modest return-wealth sorting. Greater return dispersion would amplify state dependence substantially. Figure E.3 displays the GIRFs.

**Return risk premium** The interest rate effect of return-wealth sorting is  $-0.001$  basis points at  $\sigma_e = 0.01$ —negligible. The premium grows quadratically with  $\sigma_e$  but requires  $\sigma_e \approx 0.83$  to reach 1 basis point. This identifies the amplification channel, not the pricing channel, as the economically significant consequence of return heterogeneity at realistic calibrations. The full sensitivity analysis appears in Appendix C.3.



Figure E.3: State-dependent GIRFs to a 2% negative TFP shock



*Notes:* Averaged over 1,000 stochastic TFP paths. Solid: launched from the 1952 distributional state (low inequality). Dashed: launched from the 2019 distributional state (high inequality). The dotted vertical line marks the shock arrival.

**Counterfactual analysis** With three structural shocks, I compare the baseline (all historical shocks) with three counterfactual economies—fixing  $\beta_t$ , fixing  $Z_{L,t}$ , or fixing both—to isolate each shock’s marginal contribution. Figure E.4 compares log output paths. Three findings stand out. First, fixing  $Z_{L,t}$  at its sample mean ( $\bar{Z}_L = 0.966$ ) yields a covariance share of +55 percent. Second, fixing  $\beta_t$  at its sample mean ( $\bar{\beta} = 0.967$ ) yields −24 percent: patience partially offsets output cycles. Third, fixing both (TFP only) yields a standard-deviation ratio of 0.68 against the baseline. This confirms that aggregate TFP is the primary driver of business-cycle amplitude—consistent with [Krusell and Smith \(1998\)](#)—while  $Z_{L,t}$  and  $\beta_t$  contribute meaningful but secondary variation.

TFP is the dominant driver of wages and the long-run interest rate,  $Z_{L,t}$  is the primary driver of hours fluctuations, and  $\beta_t$  partially offsets TFP-driven investment cycles through countercyclical patience.

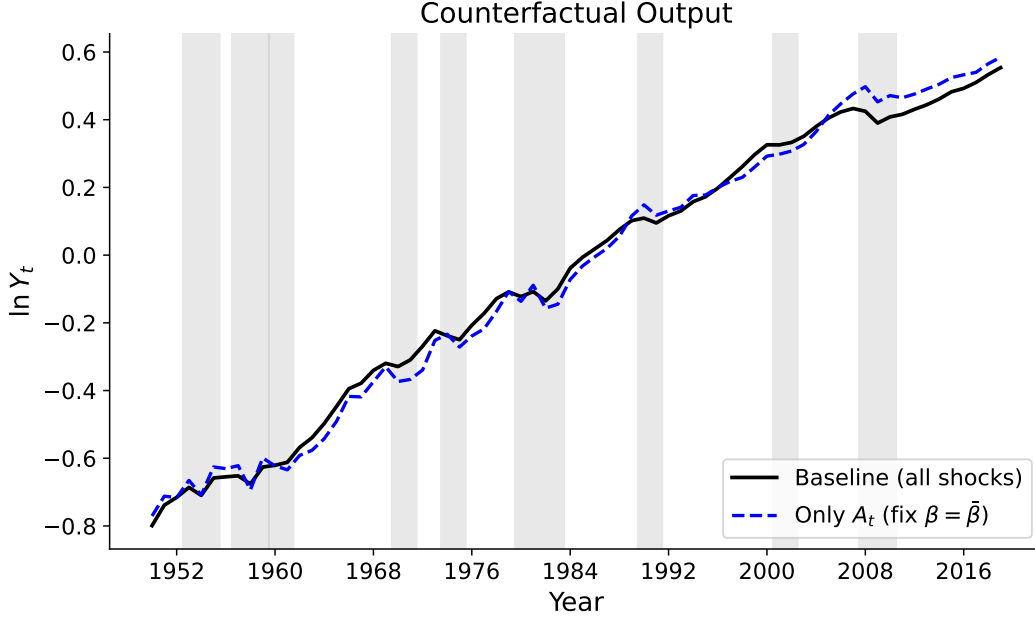
## E.4 Counterfactual multi-variable panels

Figure E.5 extends the counterfactual comparison to output, labor supply, the interest rate, and the wage.

## E.5 Transition dynamics

Figure E.6 plots the dominant eigenvalue governing distributional persistence over time. The dominant eigenvalue has mean 1.02 and exceeds unity in 100 percent of sample years, underscoring the extreme persistence of distributional dynamics. The near-unit-root behavior

Figure E.4: Counterfactual output paths



*Notes:* Baseline: all structural shocks at historical values. Fix  $\beta$ :  $\beta_t$  fixed at sample mean  $\bar{\beta} = 0.967$ , only  $(A_t, Z_{L,t})$  vary. Fix  $Z_L$ :  $Z_{L,t}$  fixed at sample mean  $\bar{Z}_L = 0.966$ , only  $(A_t, \beta_t)$  vary. Fix both: only  $A_t$  varies. Shaded areas indicate NBER recession dates.

generates endogenous persistence in aggregate dynamics that converts transitory shocks into persistent output movements.

## E.6 Distributional state dynamics

Figure E.7 plots the two covariances and their capital-normalized counterparts. From the moment-recursion inversion, the productivity-wealth covariance  $\text{cov}_t(z, a)$  increases from 0.032 in 1950 to 0.133 in 2019 (a factor of 4.2), while  $\text{cov}_t(e, a)$  grows from 0.256 to 0.891 (a factor of 3.5). Both are procyclical. The normalized ratio  $\text{cov}(e, a)/K$  remains nearly constant, reflecting the near-unit-root dynamics of  $e$ .

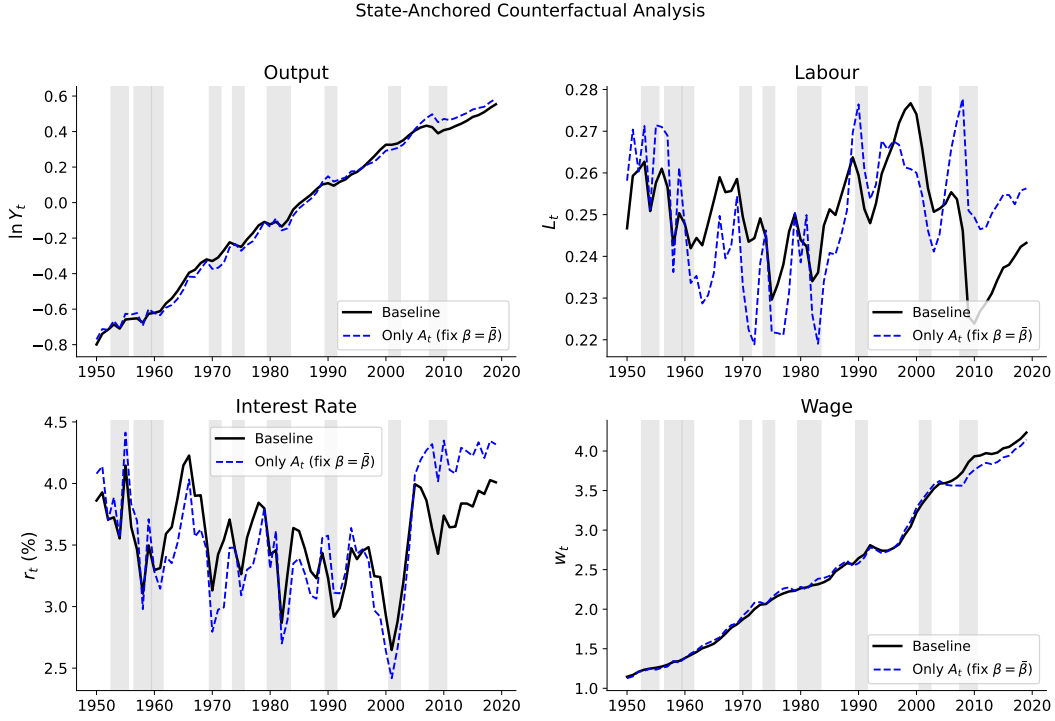
## E.7 Consumption dynamics

The MRCE delivers closed-form aggregate consumption:

$$C_t = \frac{1 - \beta_t}{1 + \theta} \left[ (1 + \tilde{r}_t + \tilde{w}_t Z_{L,t} \tilde{\xi}_t) K_t + \tilde{r}_t \text{cov}_t(e, a) + \tilde{w}_t Z_{L,t} \tilde{\xi}_t \text{cov}_t(z, a) \right] + \kappa \tilde{w}_t Z_{L,t},$$

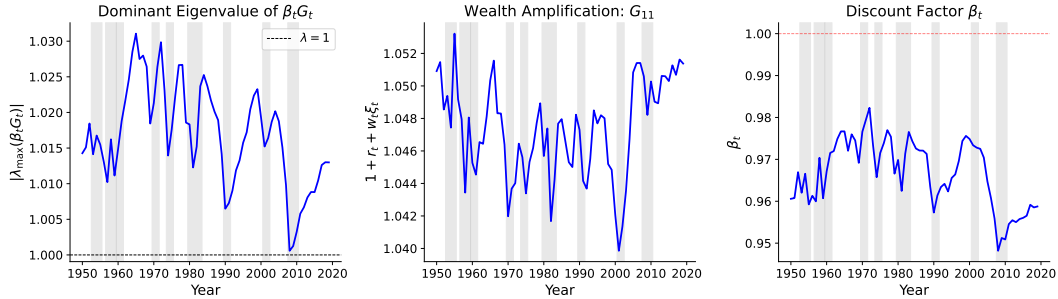
where  $\tilde{\xi}_t \equiv \xi / A_t^{1/(1-\alpha)}$  is the growth-adjusted complementarity coefficient.

Figure E.5: Counterfactual comparison across macroeconomic variables



Notes: Panels show output (log), labor supply, interest rate, and wage. Each panel compares the baseline with the fix- $\beta$  counterfactual. Shaded areas indicate NBER recession dates.

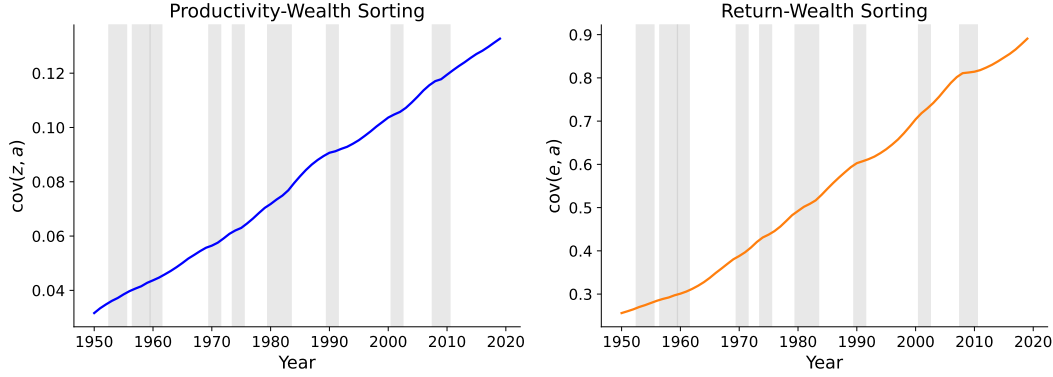
Figure E.6: Transition dynamics



Notes: Top: dominant eigenvalue governing distributional persistence. Middle: wealth amplification factor. Bottom: time-varying discount factor. Shaded areas indicate NBER recession dates.

**Constant marginal propensity to consume** The household MPC out of a one-time transfer is  $(1 - \beta_t)/(1 + \theta)$ , independent of wealth. At baseline ( $\beta = 0.96$ ,  $\theta = 4$ ), the annual MPC is 0.8 percent; under the sequential inversion, mean  $\beta_t = 0.967$  yields an average MPC of 0.665 percent. The constant MPC follows directly from linear policy functions, contrasting with models where borrowing-constrained households exhibit MPCs of 0.3–0.5 (Kaplan et al., 2018).

Figure E.7: Distributional state dynamics, 1950–2019



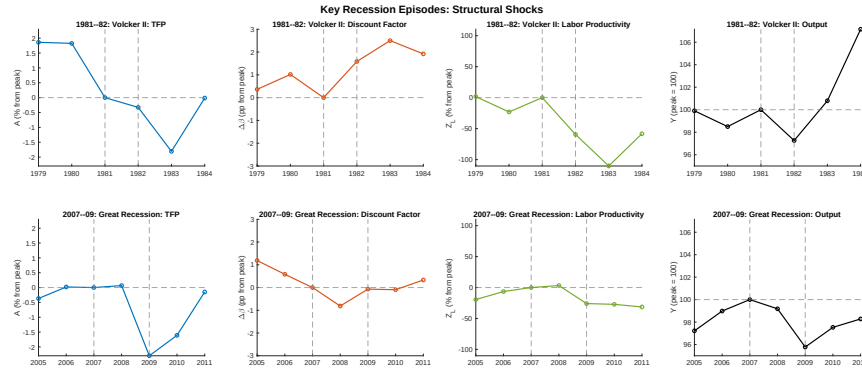
Notes: Top row: raw covariances. Bottom row: capital-normalized ratios. Shaded areas indicate NBER recession dates.

**Procyclical MPC** The time-varying MPC  $(1 - \beta_t)/(1 + \theta)$  is procyclical: when  $\beta_t$  is low (less patience), the MPC is higher. Under the sequential inversion, mean  $\beta_t = 0.967$  (std 0.0079) yields an average MPC of 0.665 percent.

## E.8 Historical episode analysis

The three-shock inversion provides a structural narrative for each recession. At annual frequency, three mild recessions (1960–61, 1969–70, 2001) produce negligible output declines and are omitted. The full classification appears in the main text (Section 5.1, Table 5). Figure E.8 shows shock paths for two contrasting episodes.

Figure E.8: Structural recession classification: shock paths around two contrasting episodes



Notes: 1981–82 (Volcker II, TFP/Labor) and 2007–09 (Great Recession, TFP-driven). Each row shows one episode. Panels show TFP ( $A_t$ , log change from peak), discount factor ( $\beta_t$ , pp from peak), labor productivity mean ( $Z_{L,t}$ , log deviation from peak), and indexed output ( $Y_t$ , peak = 100). See the main text (Table 5) for the full classification.

## E.9 Cross-sectional distributional dynamics

We simulate 500,000 agents along the historical path (1950–2019) using the three-shock inversion results. Each agent  $i$  carries state  $(a_{it}, z_{it}, e_{it})$ , where  $z$  and  $e$  follow AR(1) processes with  $\sigma_z = 0.12$ ,  $\sigma_e = 0.01$  at baseline. Countercyclical dispersion amplifies these:  $\sigma_{z,t} = 1.75 \times \sigma_z$  when  $Z_{L,t}$  declines (Storesletten et al., 2004), and  $\sigma_{e,t} = 4.1 \times \sigma_e$  when  $\Delta Y_t < 0$  (Bloom et al., 2018). Wealth evolves as  $a_{it+1} = \beta_t R_{it} a_{it} + T_t$  where  $R_{it} = 1 + \tilde{r}_t e_{it} + \tilde{w}_t Z_{L,t} \tilde{\xi}_t z_{it}$ , and consumption is  $c_{it} = \frac{1-\beta_t}{1+\theta} R_{it} a_{it} + \kappa \tilde{w}_t Z_{L,t} z_{it}$ .

The simulation begins after a 2,000-period burn-in at steady-state prices with GE feedback (recomputing  $r$ ,  $w$  from  $K_t = \mathbb{E}[a_{it}]$  each period) and wealth truncation at  $\bar{a} = 200$ . Return dispersion  $\sigma_e$  ramps gradually from 30% to 100% of its calibrated value during the burn-in, reflecting the postwar era’s compressed return heterogeneity. Post-burn-in wealth Gini is approximately 0.76. During the historical simulation, the truncation boundary scales with capital ( $\bar{a}_t = (\bar{a}/K_{ss}) \cdot K_t$ ) to maintain a constant max-to-mean ratio.

Distributional moments by recession appear in the main text (Table 9). Two patterns emerge: (i) consumption inequality rises modestly in recessions (CV change +3.49%) while income CV falls (−5.42%); (ii) the Great Recession produces the highest consumption dispersion (CV 1.28, +26.5% above normal), reflecting falling  $\beta_t$  interacting with high wealth concentration.

## E.10 Robustness

Table E.1 reports sensitivity to alternative calibrations. Approximate aggregation is extremely robust:  $I^{\text{rep}}$  accounts for 93–99 percent of total investment across all configurations. Near-unit-root persistence is robust to all parameters except the transfer share  $\eta$ .

## F Micro inversion: data construction

This appendix documents the data construction underlying the household inversion of Section 4.2 and the household-level analyses in Sections 5.5–5.7. Two surveys are used. PSID 2005–2019 (8 biennial waves) supplies the panel structure required by the saving-equation identification. SCF 1992–2019 (10 triennial waves) supplies the cross-sectional intratemporal-FOC identification; the 1989 wave is excluded as a methodology outlier ( $n=1,067$  vs. 11,000–18,000 in subsequent waves;  $\rho(\beta_i, \log a_i)$  sign-flips against all later waves due to first-survey weighting that the AHEAD redesign fixed in 1992). All household-level variables are deflated to 2021 dollars using the CPI-U annual average and converted to model units by dividing

Table E.1: Sensitivity of steady-state properties to alternative calibrations

| Parameter       | Value | $K/Y$ | $r$ (%) | $\lambda_{\max}$ | $I^{\text{rep}}$ (%) | Half-life (yr) |
|-----------------|-------|-------|---------|------------------|----------------------|----------------|
| <i>Baseline</i> |       | 3.38  | 2.77    | 0.9992           | 94.4                 | 827            |
| $\alpha$        | 0.3   | 3.02  | 2.93    | 0.9991           | 94.0                 | 749            |
|                 | 0.36  | 3.77  | 2.54    | 0.9992           | 94.8                 | 905            |
|                 | 0.4   | 4.41  | 2.07    | 0.9993           | 95.7                 | 1001           |
| $\sigma_e$      | 0.001 | 3.24  | 3.17    | 0.9988           | 97.8                 | 573            |
|                 | 0.005 | 3.29  | 3.02    | 0.9990           | 96.5                 | 695            |
|                 | 0.02  | 3.54  | 2.32    | 0.9993           | 90.7                 | 973            |
|                 | 0.05  | 3.86  | 1.54    | 0.9994           | 84.3                 | 1157           |
|                 | 0.1   | 4.13  | 0.98    | 0.9995           | 79.9                 | 1276           |
| $\theta$        | 1     | 3.38  | 2.77    | 0.9992           | 94.4                 | 827            |
|                 | 1.5   | 3.38  | 2.77    | 0.9992           | 94.4                 | 827            |
|                 | 3     | 3.38  | 2.77    | 0.9992           | 94.4                 | 827            |
|                 | 5     | 3.38  | 2.77    | 0.9992           | 94.4                 | 827            |
| $\eta$          | 0.005 | 3.35  | 2.84    | 0.9997           | 95.0                 | 2251           |
|                 | 0.01  | 3.37  | 2.79    | 0.9994           | 94.6                 | 1110           |
|                 | 0.02  | 3.40  | 2.70    | 0.9987           | 93.8                 | 541            |
|                 | 0.03  | 3.43  | 2.61    | 0.9980           | 93.1                 | 353            |
|                 | 0.05  | 3.51  | 2.41    | 0.9966           | 91.4                 | 204            |
| $\beta$         | 0.94  | 2.62  | 5.62    | 0.9991           | 88.6                 | 738            |
|                 | 0.95  | 2.95  | 4.19    | 0.9991           | 91.5                 | 790            |
|                 | 0.97  | 3.98  | 1.29    | 0.9992           | 97.0                 | 816            |
|                 | 0.98  | 5.03  | -0.44   | 0.9992           | 98.1                 | 903            |
| $\rho_z$        | 0.9   | 3.37  | 2.79    | 0.9992           | 94.5                 | 837            |
|                 | 0.93  | 3.37  | 2.78    | 0.9992           | 94.5                 | 834            |
|                 | 0.97  | 3.40  | 2.71    | 0.9991           | 93.9                 | 808            |
|                 | 0.99  | 3.65  | 2.04    | 0.9991           | 88.4                 | 794            |

*Notes:* Each row varies one parameter from the baseline calibration.

by  $\text{dpm} \equiv 34,862 \times (\text{CPI}_{2021}/\text{CPI}_{2017}) \approx 38,545$ . The factor matches the macro inversion's per-capita 2017-dollar normalization.

## F.1 Common conventions

**Time endowment.** The labor share  $n_i$  is computed against an annual discretionary endowment of  $17 \text{ h/day} \times 365 \text{ d/yr} = 6,205$  hours per worker. Households with multiple earners use the per-earner endowment scaled by the number of earners. The choice of 17 hours per day matches the time-use literature's residual after sleep and personal care, and matches the calibration in the macro side  $(\kappa, \xi)$ .

**Macro GE inputs.** For each survey wave, we read  $(A_t, \beta_t^{\text{macro}}, Z_{L,t}, r_t, w_t, K_t, L_t)$  from the macro inversion’s annual output (main text Section 4). After-tax prices are  $\tilde{r}_t = r_t(1 - \tau_K)$  and  $\tilde{w}_t = w_t(1 - \tau_L)$  at the calibrated  $\tau_K = \tau_L = 0.30$ . The transfer is  $T_t = \eta\tau Y_t$  with  $\eta = 0.0133$ . The growth-adjusted complementarity is  $\tilde{\xi}_t = \xi/A_t^{1/(1-\alpha)}$  with  $\xi = 0.0213$ , where  $\alpha$  is the Penn World Tables capital share for year  $t$ .

**Renormalization of  $z_i$ .** Within each wave, we renormalize  $z_i$  so that  $E_w[z_i \mid \text{worker}] = 1$ . The macro per-capita wage  $w_t$  averages over the total population (workers and non-workers); the survey conditions on workers. The renormalization absorbs this universe-vs-sample wage-level gap so that cross-sectional dispersion of  $z_i$  is the relevant heterogeneity object.

**Renormalization of  $e_i$ .** We compute  $e_i^{\text{raw}} = \text{KI}_i/(r_t a_i)$  using the gross (pre-tax) macro return  $r_t$  and PSID/SCF-reported  $\text{KI}_i$  in model units. Within each wave, we renormalize so that  $E_w[e_i \mid \text{KI}_i > 0] = 1$ , which absorbs both the gross-vs-net-tax convention  $(1 - \tau_K)$  and any sample-level reporting biases in capital-income variables. Households with  $\text{KI}_i^d \leq 0$  have  $e_i$  imputed to 1, on the interpretation that PSID/SCF capital-income items (interest, dividends) miss housing equity and retirement-account returns.

## F.2 PSID biennial saving-equation derivation

The PSID identification rests on a closed-form quadratic in  $\beta_i$  obtained by iterating the one-period saving rule through two consecutive years to match PSID’s biennial wave structure. Starting from the household policy

$$a_{i,t+1} = \beta_i a_{i,t} R_{i,t} + T_t, \quad R_{i,t} \equiv 1 + \tilde{r}_t e_i + \tilde{w}_t Z_{L,t} \tilde{\xi}_t z_i,$$

and applying it again from  $t + 1$  to  $t + 2$  under the (BRCE) assumption that the household treats  $\beta_i$  and the structural parameters as time-invariant, we obtain

$$\begin{aligned} a_{i,t+2} &= \beta_i a_{i,t+1} R_{i,t+1} + T_{t+1} \\ &= \beta_i [\beta_i a_{i,t} R_{i,t} + T_t] R_{i,t+1} + T_{t+1} \\ &= \beta_i^2 a_{i,t} R_{i,t} R_{i,t+1} + \beta_i T_t R_{i,t+1} + T_{t+1}. \end{aligned}$$

Rearranged as a quadratic in  $\beta_i$  with given  $(a_{i,t}, a_{i,t+2}, R_{i,t}, R_{i,t+1}, T_t, T_{t+1})$ ,

$$[a_{i,t} R_{i,t} R_{i,t+1}] \beta_i^2 + [T_t R_{i,t+1}] \beta_i + [T_{t+1} - a_{i,t+2}] = 0. \quad (21)$$

Let  $\mathcal{D} \equiv T_t^2 R_{i,t+1}^2 + 4a_{i,t} R_{i,t} R_{i,t+1} (a_{i,t+2} - T_{t+1})$  denote the discriminant.  $\mathcal{D} > 0$  whenever  $a_{i,t+2} > T_{t+1}$  (next-wave wealth strictly exceeds the transfer floor), which holds for every household in the consistent estimation sample. The smooth positive root

$$\beta_i = \frac{-T_t R_{i,t+1} + \sqrt{\mathcal{D}}}{2a_{i,t} R_{i,t} R_{i,t+1}}$$

is the recovered preference parameter; the negative root is discarded as economically meaningless. The main-text exposition in Section 4.2 and the implementation both collapse to the constant-price form

$$a_{i,t+2} = \beta_i^2 a_{i,t} R_{i,t}^2 + T_t (1 + \beta_i R_{i,t}),$$

which assumes  $R_{i,t} = R_{i,t+1}$  and  $T_t = T_{t+1}$  within the wave-pair window. This matches the belief-invariance assumption: households treat current prices and the transfer as time-invariant when forming saving decisions. Over a biennial window with prices and transfers that move only modestly, the collapsed form is the natural empirical counterpart to the household’s perceived environment.

### F.3 PSID variables and sample

The PSID processed panel underlying the inversion is built from `psid_family_panel.rds` and combines biennial family-level files 2005–2019 (`family_id`, `wave_year`, `pid`). The R pipeline that produced the processed CSV lives in `empirics_psid/code/02_construct_vars_and_estimates` and the Python inversion in `exact_ge_mapping/micro_inversion.py` reads the resulting CSV directly. Key variables:

- $W_{\text{nom},i,t} = \text{Wd\_nom}$ : pre-tax full-time-equivalent annual wage, computed as labor income divided by  $n_i$  (the household’s labor share against the per-earner endowment).
- $a_{i,t} = \text{a\_m}$ : net worth (PSID `net_worth_total`) deflated to 2021\$ and converted to model units.  $a_{i,t+2} = \text{a\_prime\_m}$  is the same variable two waves ahead, deflated by the next-wave CPI and converted to model units.
- $\text{KI}_{i,t} = \text{capital\_income\_real}$ : PSID’s reported sum of interest income, dividend income, business income, and net rental income, deflated to 2021\$.
- Survey weight = `weight_reweighted`: the PSID longitudinal family weight, renormalized within the consistent estimation sample so weights sum to the original total.
- `age_head`, `race_head`, `educ_head`: demographics on the family head, used in lifecycle and conditional analyses.



The estimation sample restricts to households satisfying `in_consistent = TRUE`: positive after-tax wage, positive labor share, positive wealth in both current and next wave, valid biennial gap, and positive longitudinal weight. Sample sizes range from 3,864 (2013) to 4,372 (2005).

## F.4 SCF variables and sample

SCF data are read directly from the Federal Reserve summary extracts (`rscfpYYYY.dta`) and full extracts (`pYYi6.dta`, with `p92i4.dta` as the 1992 exception). Hours data are merged from the full extract:

- `x4110` = hours per normal week,
- `x4111` = weeks per normal year,
- `x4507` = hours per week at second job.

Pre-2013 SCF full extracts use uppercase column names (`X4110`, etc.); 2013 onwards uses lowercase. The Python loader normalizes case before extraction.

SCF reports five implicates per household to handle imputation; the survey weight `wgt` accounts for this (each implicate has weight = household weight / 5). Key variables:

- $a_{i,t}$  = `networth` deflated to 2021\$ then divided by `dpm`.
- $W_{\text{nom},i,t}$  = `wageinc` divided by the household's labor share  $n_{i,t}$  (when  $n_{i,t} > 0.01$ ).
- $KI_{i,t}$  = `intdivinc` (interest + dividends) + `bussefarminc` (business and farm income) + `kginc` (capital gains), all deflated to 2021\$. SCF's broader capital-income coverage is one reason its  $e_i$  identification is tighter than PSID's.
- Survey weight = `wgt`.
- `age`, `edcl`, `race`, `married`: demographics.

The estimation sample restricts to households with positive net worth, positive labor share, and positive weight. We further restrict to working households ( $n_i > 0.01$  and  $W_{\text{nom},i} > 0$ ) for the intratemporal-FOC identification. Non-workers'  $\beta_i$  would require the consumption equation directly (with  $z_i = 0$ ), and consumption data quality in SCF is uneven; we leave non-workers for future work.

## F.5 $\beta$ -clipping and quality flags (SCF only)

The SCF intratemporal-FOC formula

$$\beta_i = 1 - \frac{(1 + \theta) \tilde{w}_t Z_{L,t} z_i (\kappa + \tilde{\xi}_t a_i - n_i)}{\theta R_i a_i}$$

is sensitive to whether the implied leisure quantity  $\kappa + \tilde{\xi}_t a_i - n_i$  is positive. For low-wealth households whose hours  $n_i$  exceed  $\kappa + \tilde{\xi}_t a_i$  (the model-implied time endowment), the bracket flips sign and  $\beta_i$  exceeds unity. For numerical stability we clip  $\beta_i$  at  $[0.5, 1.5]$ . The clipping affects 1–10% of SCF households per wave (mostly very-low-wealth full-time workers); the clip indicator is reported as `beta_clipped` in the per-household output for transparency. PSID does not require clipping because the saving-equation quadratic admits a smooth positive root for any  $a_{i,t+2} > T_t$ .

## F.6 Output files and reproducibility

The Python module `exact_ge_mapping/micro_inversion.py` produces:

- `csv_output/micro_inversion_psid.csv`: per-household-per-wave  $(z_i, e_i, \beta_i, R_i)$  for PSID, with demographics, weight, and the imputation indicator.
- `csv_output/micro_inversion_psid_summary.csv`: per-wave aggregate moments (sample sizes,  $\beta_{\text{cap}}, \beta_{\text{pop}}, \beta_{\text{med}}$ , all relevant correlations,  $\widehat{\text{cov}}_t(z, a)$  and  $\widehat{\text{cov}}_t(e, a)$ ).
- `csv_output/micro_inversion_scf.csv` and `micro_inversion_scf_summary.csv`: same structure for SCF.
- `csv_output/micro_lifecycle.csv`: 5-year-bin lifecycle profiles pooled across waves, with both surveys.
- `figures/micro_*.pdf`: six diagnostic figures (distributions by wave, scatter against  $\log a$ , correlation stability, PSID-vs-SCF  $\beta_{\text{cap}}$ , lifecycle profiles,  $\beta$ -macro-vs-micro time series).

The companion module `exact_ge_mapping/micro_diagnostics.py` reads the panel CSVs and produces the per-table CSV outputs used in Sections 5.5, 5.6, and 5.7:

- `micro_diag_vardecomp.csv`: variance decomposition of  $\log z$  and  $\beta$  into person FE / age trend / wave / idiosyncratic (Table 12).

- `micro_diag_policy_invariance.csv`: pre/post wave-level moments around the four major federal tax reforms (Table 14).
- `micro_diag_delta_beta.csv`: within-person biennial  $\Delta\beta_i$  statistics by wave-pair plus the pooled noise band (Section 5.6).
- `micro_diag_age_corr.csv`: age-bin Pearson correlations  $\rho(\beta, z)$ ,  $\rho(\beta, \log a)$ ,  $\rho(z, \log a)$  for both surveys (Table 13).
- `micro_quartile_table.csv`: within-survey wealth-quartile decomposition of  $a, \beta, \beta_{\text{cap}}, z$  (Table 11); unweighted  $a$  and  $\beta$  medians within Q4 to match the SCF dealer-sample convention.
- `micro_diag_fin_wealth.csv`: PSID inversion using `financial_wealth_real` (excluding housing equity) instead of total net worth (financial-wealth robustness footnote in Section 5.7).

The modules are invoked from `main.py` after the macro inversion completes; the macro CSV `task2e_ABZL_inversion.csv` is the only macro-side input. Re-running the macro inversion regenerates the GE inputs; re-running the micro inversion and diagnostics against an updated macro CSV requires no manual steps. PSID and SCF processed CSVs are read read-only from a sibling project directory.

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